

1. Example 4b from §5.2 Day 1 is an example of a *telescoping series* which means

$$(\#4b) \sum_{n=1}^{\infty} \frac{1}{n(n+1)}$$

2. (★) A **geometric series** has form

3. For each series below, determine whether or not they converge. (Justify your conclusion.) If they do converge, determine their sum.

$$(a) \sum_{n=1}^{\infty} \pi \left(\frac{2}{3} \right)^{n-1}$$

$$(b) \sum_{n=1}^{\infty} \frac{4^{n-1}}{3^{n+1}}$$

$$(c) \sum_{n=1}^{\infty} 10 \left(\frac{-3}{5} \right)^n$$

4. (★) The **harmonic series** is

5. Extra Practice

$$(a) \sum_{n=1}^{\infty} (e^{2/n} - e^{2/(n+1)})$$

$$(b) \sum_{n=1}^{\infty} \left[\left(\frac{-1}{2} \right)^{n-1} + \left(\frac{4}{5} \right)^n \right]$$